HH–C–581a

DECEMBER 18, 1935

SUPERSEDING
Fed. Spec. HH–C–581
September 5, 1933

FEDERAL SPECIFICATION

COTTON FABRIC; WOVEN, ASPHALT-SATURATED

This specification was approved by the Director of Procurement, for the use of all departments and establishments of the Government, and shall become effective not later than July 15, 1936. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE FEDERAL SPECIFICATIONS.

A–1. Federal Specification CCC–T–191—Textiles; Test Methods—of the issue in effect on date of invitation for bids, shall form a part of this specification.

A–2. Any special requirements of the individual departments of the Government are noted under section H.

B. TYPES.

B–1. Asphalt-saturated woven cotton fabric shall be of but one type.

C. MATERIAL.

C–1. Material shall be composed of woven cotton fabric impregnated with an asphaltic saturant.

D. GENERAL REQUIREMENTS.

D–1. See Detail Requirements.

E. DETAIL REQUIREMENTS.

E–1. *Appearance.*—It must be free from visible external defects, such as ragged or untrue edges, breaks, and rents. The meshes of the fabric shall not be completely closed or sealed by the asphalt saturant. There shall be sufficient porosity to allow the moppings of asphalt plying cement to come through. The fabric shall not stick when unrolled at temperatures between 10° C. (50° F.) and 32° C. (90° F.) to such an extent as to cause injury. The surface of the fabric shall be uniformly smooth and free from irregularities, folds, and knots. It shall not be coated with talc, portland cement, or other substance which would tend to interfere with the adhesion between the fabric

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F–3. *Sampling for laboratory examination.*—From the rolls examined for sampling, select one the weight of which is near the average weight of the whole lot. Unroll and at approximately 25 feet from the end cut two samples the full width of the roll and approximately 40 inches long, label, carefully wrap, and transmit one to the testing laboratory. Retain the other sample for check analysis in case of dispute.

F–4. *Laboratory examination.*—

F–4a. *Appearance.*—Examine both sides of the sheet of saturated fabric and note the color and general appearance.

F–4b. *Weight.*—Trim the sample so that it is 36 inches long and the full width of the roll, measure accurately to one-sixteenth inch, and weigh to the nearest gram (15 grains). From the measurements and weight so obtained compute the weight per square yard. The weight so determined is only to be used in making the laboratory calculations and shall not be used as a basis for acceptance or rejection. (See F–2.)

F–4c. *Pliability.*—Cut five strips 1 inch wide and 6 inches long in the direction of the warp and immerse in a cooling mixture of ice and water at 0° C. (32° F.) for 10 to 15 minutes. The strips shall then be removed from the water and each strip bent over a 1/32-inch mandrel through an arc of 180° at a uniform speed in approximately 2 seconds and then through 360° over the same mandrel in the opposite direction. Dry thoroughly and examine. If 1 or more of these 5 strips crack on this test, cut 10 strips from another part of the sample and repeat the test. If one or more of these strips crack the material is reported as failing on the pliability test.

F–4d. *Breaking strength.*—(Grab method.) Follow method given in section VI, Federal Specification CCC–T–191.

F–4e. *Volatile matter.*—Cut two strips of asphalt-saturated fabric 6 inches wide and 12 inches long. Weigh the strips separately and suspend them for 5 hours in an oven at 105° to 110° C. (221° to 230° F.), remove from the oven, cool in a desiccator, and weigh. Compute the percentage loss in weight and average the two results.

F–4f. *Weight of desaturated fabric.*—Cut a strip approximately 2 inches wide and the full width of the sample, weigh, and extract with benzol (carbon tetrachloride, chloroform, or carbon bisulphide will usually give the same results but benzol should be used in case of dispute) in a suitable extraction apparatus until the solvent runs through colorless. The saturated fabric may be cut into squares approximately 1 inch on each side in case an extraction apparatus of sufficient size to accommodate the whole strip is not available. Remove the desaturated fabric from the extraction apparatus, dry in air until free from the odor of benzol, and brush off any adhering particles of foreign matter. Place the desaturated fabric in a tared weighing bottle and heat in an oven at 105° to 110° C. (221° to 230° F.) for one half hour, cool in a desiccator and weigh. Calculate the percentage of moisture-free fabric in the sample taken for extraction and compute the weight of moisture-free desaturated fabric per square yard from the weight obtained in F–4b.

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and the plying cement. The use of powdered silica or wood flour shall be permitted.

E–2. *Physical and chemical requirements.*—

	Maximum	Minimum
Width.....inches..	38	30
Selvage.....inch..	¼	
Weight per roll. To be mutually agreed upon by buyer and seller but in no case more than 80 pounds.		
Weight per square yard.....ounces..	15	11
Average strength (grab method):		
Direction of the warp.....pounds..		50
Direction of the filling.....do....		40
Volatile matter at 105° C. (221° F.).....percent..	2	
Pliability at 0° C. (32° F.). (Cracking on bending over a ½₂-inch mandrel through an arc of 180° in one direction and then through 360° over the same mandrel in the opposite direction).....		None
Average weight of moisture-free desaturated fabric per square yard ounces..		3½
Saturation of moisture-free fabric.....percent..		200
Ash of moisture-free desaturated fabric.....do....	1	
Thread count per inch. (Both in the direction of the warp and the filling).....	36	18
Cotton fiber content.....percent..		[1] 100

[1] This shall not exclude the customary sizing materials in amounts not exceeding 8 percent by weight of the moisture-free cotton.

F. METHODS OF INSPECTION, DETERMINATION OF WEIGHTS AND WIDTH SAMPLING AND TESTING.

Deliveries will, in general, be sampled and tested by the following methods, but the purchaser reserves the right to use any additional information to ascertain whether the material meets the specification.

F–1. *Sampling for inspection.*—From each shipment of more than 1,000 rolls taken at random a number of rolls of fabric equivalent to one-half the cube root of the total included in the lot. If this proves to be a fractional number, express it as the next higher digit. From each shipment of 1,000 rolls or less take at random 5 in each case.

F–2. *Determination of weight and width.*—The weight per square yard of material, and its variation in weight; the width and its variation must be determined by the inspector and the results transmitted to the testing laboratory with the sample. Material which does not conform to the specification in these respects should be rejected by the inspector and only deliveries which conform in these respects sampled for laboratory tests.

F–2a. Remove wrappers and other packing material from each roll selected and weigh the roll and the packing materials, etc., separately to the nearest one-quarter pound, then unwind each roll and measure the length and width to the nearest one-quarter inch and observe the appearance of the material. (See E–1.) Compute the area of each roll and the weight per square yard of each roll of fabric examined.

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F–4g. *Saturation of moisture-free fabric.*—The weight of saturant in the fabric is the difference in weight between the original sample taken for extraction in F–4f and the weight of the desaturated fabric (plus any surfacing material present). The percentage saturation of the moisture-free fabric is obtained by the following formula:

$$\frac{\text{Weight of saturant}}{\text{Weight of moisture-free desaturated fabric}} \times 100 = \text{percentage saturation}$$

F–4h. *Thread count.*—Count the number of threads to the inch in both the warp and filling at five different places. The average of these shall be taken as the threads per inch or thread count.

F–4i. *Character of fiber.*—From a piece of the desaturated fabric (from which any sizing has been removed by bringing to a boil in a 0.5 percent sodium hydroxide solution), take at random 5 threads in the direction of the warp and 5 in the direction of the filling and separate the fibers in water on a microscope slide, so that the fibers are uniformly distributed. Dry the slides so prepared until all the water is removed, cool and add a drop of zinc-chloriodide solution (Herzberg solution).[1]

Cover the stained fibers with a cover glass, draw off the excess stain, and examine under a microscope with a magnification of about 100 diameters. The fibers should all be stained a wine-red color, and shall have the characteristic shape of cotton fiber.

G. PACKAGING, PACKING, AND MARKING.

G–1. *Packaging.*—Unless otherwise specified, commercial packages are acceptable under this specification.

G–2. *Packing.*—

G–2a. The rolls of treated cotton fabric shall be wound on mandrels[2] not less than 2 inches in diameter and extending 2 to 4 inches beyond the end of the rolls. They shall be securely tied or wrapped in a manner which will prevent unrolling in transit.

G–3. *Marking.*—Each package shall be legibly marked with a description of the contents, the name of the manufacturer, the place of manufacture, the brand (if any), the net weight of the contents, and some means of identification of "the particular contract on which the purchase and shipment were made." Each shipment of the material shall be accompanied by printed directions for its use.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS.

H–1. The following departmental specifications, of the issue in effect on date of invitation for bids, shall respectively form a part of this specification.

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[1] The Herzberg stain is made according to the following formula:

Solution A { 20 g zinc chloride,
10 ml distilled water.

Solution B { 2.1 g potassium iodide,
0.1 g iodine,
5.0 ml distilled water.

Dissolve solutions A and B separately, then mix and allow to stand several hours, or until all sediment has settled out. The clear liquid is then decanted into a dark-colored bottle and is ready for use. The stain shall be kept in the dark and should be tested before use on known fibers.

[2] The mandrel may be either round, square or polygonal.

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H–1a. *Army.*—United States Army Specification No. 100–2, Standard Specification for Marking Shipments.

H–1b. *Navy.*—Navy Department purchases will be made under the issue in effect on date of invitation for bids of Bureau of Yards and Docks Specification No. 7Y, Roofing, Siding, and Sheet Metal Work; Dampproofing and Membrane Waterproofing, copies of which may be obtained without cost upon application to the Bureau of Yards and Docks, Navy Department, Washington, D. C.

I. NOTES.

I–1. Basis of purchase shall be by the roll or by the square (100 square feet) according to agreement between purchaser and the seller.

I–2. Asphalt-saturated fabric is intended for use with asphalt for waterproofing and dampproofing in the construction of membrane waterproofing.

I–3. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bid, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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